I developed a machine learning pipeline to predict freight rates using shipment distance, weight, equipment type, geographic routes, and market indicators.

I experimented with different regression models, engineered additional features, and handled data quality issues to improve prediction accuracy. My final solution combines HistGradientBoosting and LightGBM, using rate per mile as the prediction target.

I analyzed 48,000 labeled records and identified several data quality issues:

- Negative weights: Corrected 292 records using absolute values.
- Missing market indices: Imputed 374 values using daily medians.
- Extreme rates: Applied winsorization at the 0.5th and 99.5th percentiles to reduce the influence of outliers.

I engineered features such as weight per mile, log distance, Haversine distance, circuity, geographic coordinates, and temporal indicators.

I compared multiple regression models and selected an ensemble of HistGradientBoosting and LightGBM. Predicting rate per mile helped the models capture pricing patterns beyond shipment distance.

I used expanding-window rolling-origin cross-validation across four months to evaluate performance on future data.

Metric	**Final Ensemble**
MAE (All Loads)	$118.41
MAE (Clean Loads)	$64.85
R^2 (Clean Loads)	0.9943
MAPE (All Loads)	5.14%

The ensemble outperformed the baseline models in the reported evaluation.

I generated 12,000 validation predictions and 31 daily predictions for the Lexington–Fort Wayne corridor. I recovered missing geographic coordinates using historical hub data and incorporated the available daily market indicators.

The generated files passed the schema, non-null, and positive-rate checks performed by the provided `score.py` script.

